scientific reports

Check for updates

OPEN

Artificial intelligence-based non-invasive bilirubin prediction for neonatal jaundice using 1D convolutional neural network

Fatemeh Makhloughi

Neonatal jaundice, characterized by elevated bilirubin levels causing yellow discoloration of the skin and eyes in newborns, is a critical condition requiring accurate and timely diagnosis. This study proposes a novel approach using 1D Convolutional Neural Networks (1DCNN) for estimating bilirubin levels from RGB, HSV, LAB, and YCbCr color channels extracted from infant images. Initially, each color channel is treated as a time series input to a 1DCNN model, facilitating bilirubin level prediction through regression analysis. Subsequently, RGB feature maps are combined with those derived from HSV, LAB, and YCbCr channels to enhance prediction performance. The effectiveness of these methods is evaluated based on Root Mean Squared Error (RMSE), R-squared (R^2), and Mean Absolute Error (MAE). Additionally, the best-performing model is adapted for classification of jaundice status. The results show that the integration of RGB and HSV color spaces yields the best performance, with an RMSE of 1.13 and an R^2 score of 0.91. Moreover, the model achieved an impressive accuracy of 96.87% in classifying jaundice status into three categories. This study provides a promising non-invasive alternative for neonatal jaundice detection, potentially improving early diagnosis and management in clinical settings.

Keywords Neonatal jaundice, Bilirubin level prediction, Image processing, One dimensional convolutional neural network

Neonatal jaundice is a prevalent condition in newborns, manifesting primarily due to alterations in red blood cell metabolism during the first week of life[1]. It affects approximately 60% of full-term and 80% of premature infants, with 10% of breastfed babies experiencing jaundice for up to four weeks[2,3]. The transition from fetal to neonatal life imposes a significant burden on the infant's immature system, particularly in managing the excess hemoglobin generated during this period. Shorter red blood cell lifespan in newborns, combined with conditions such as birth-related bruising or prematurity, leads to an increased production of bilirubin. This, alongside the immaturity of the liver's metabolic pathways, explains why approximately half of all neonates develop jaundice within their first week of life, with the incidence being higher among premature infants[4].

The condition usually starts at the face and upper body, gradually spreading down to the toes, and is more common in premature infants[3]. Recent studies highlight the critical need for accurate diagnosis and effective management of neonatal jaundice[5]. Since more than half of newborns develop jaundice and their immune systems are underdeveloped, timely detection is crucial to avoid severe and potentially life-threatening complications[6]. Unfortunately, in many regions, limited medical resources and financial constraints hinder regular examinations, contributing to higher mortality rates[7]. Jaundice remains a significant cause of neonatal mortality worldwide, ranking seventh in the leading causes of death within the first six days of life[8].

In Iran, neonatal jaundice is widespread, and its management poses considerable challenges. The recommended follow-up visits post-early discharge are often impractical, leading to delays in diagnosis and treatment[9]. This delay may lead to prolonged hospital stays, phototherapy, blood transfusions, and severe complications such as kernicterus, cognitive impairments, hearing loss, and reduced IQ[10–12]. Visual assessment, the most common method used in outpatient settings, including homes and clinics lacking advanced technology, is not reliably accurate[13]. Studies indicate that even experienced physicians face challenges in accurately assessing the severity of jaundice through visual observation[14].

Department of Biomedical Engineering, Imam Reza International University, Mashhad, Iran. email: fatemeh.makhloughi@yahoo.com

The diagnosis of neonatal jaundice is predominantly conducted in hospitals and clinics through the measurement of total serum bilirubin (TSB) obtained from a blood sample. TSB is widely regarded as the gold standard for bilirubin assessment due to its accuracy and reliability in quantifying bilirubin levels in the bloodstream[15,16]. However, its invasive nature presents significant challenges, particularly for newborns. The procedure involves blood draws, which can cause pain, increase the risk of infections and typically requires several hours to obtain results[17,18]. These drawbacks highlight the need for less invasive and more accessible diagnostic alternatives. A commonly used non-invasive method involves transcutaneous bilirubin (TCB) meters, such as BiliCheck[19] and JM-103[20]. These devices estimate bilirubin levels by analyzing yellow pigmentation on the skin, which correlates with bilirubin concentration in the underlying tissues. TCB meters leverage optical technologies, such as spectral reflectance or multi-wavelength analysis, to provide quick and painless bilirubin measurements. Their non-invasive nature reduces the physical and psychological stress on infants and caregivers, making them a valuable tool in neonatal care. Despite these advantages, TCB meters have notable limitations that impact their accuracy and utility. Their reliability decreases in infants with darker skin tones due to the interference of melanin with the optical signal, leading to potential underestimation or overestimation of bilirubin levels[21]. Furthermore, TCB devices become less accurate at higher bilirubin concentrations, typically above 14 mg/dL, where precise measurements are critical for determining the need for therapeutic interventions such as phototherapy or exchange transfusion[22]. Another significant limitation is the restricted availability of TCB meters. These devices are predominantly used in clinical settings and require trained healthcare professionals for operation, limiting their accessibility in resource-limited or remote areas. High costs associated with TCB devices further hinder their widespread adoption, particularly in low- and middle-income countries where neonatal jaundice remains a leading cause of morbidity and mortality. Given these challenges, recent research has explored innovative solutions to improve the diagnosis and management of neonatal jaundice[15]. Advances in smartphone-based applications and portable diagnostic tools utilizing image processing and artificial intelligence (AI) have shown promise in overcoming the accessibility and cost barriers associated with traditional methods[23]. Such technologies offer the potential for remote monitoring and early detection of jaundice, particularly in underprivileged regions where timely diagnosis is often delayed due to limited medical resources.

Given that elevated bilirubin levels manifest as yellow discoloration in the skin and eyes, image analysis has been explored for diagnosing jaundice. Recent advancements in image processing and machine learning have facilitated the development of automated jaundice detection methods[24]. For instance, Aydin et al.[25] employed artificial intelligence to classify jaundice in 80 infant images captured via smartphone. Their method involved image segmentation for color balance calibration using an 8-colored card, followed by color map transformation and feature extraction in multiple color spaces (RGB, YCrCb, LAB). They applied k-NN and support vector regression models to estimate bilirubin levels, achieving a success rate of 85% in detecting jaundice and a compliance rate of 85% with bilirubin measurements obtained from standard blood tests[25]. Munkholm et al.[26] developed a technique using the iPhone 6 camera for transcutaneous bilirubin measurement in neonates. By using a dermatoscope and a Wratten No. 11 filter, they enhanced bilirubin detection. In their study of 64 healthy Caucasian newborns, green and blue color intensities from images correlated significantly with bilirubin levels ($p < 0.001$), with Pearson's coefficients of 0.59 and 0.48, respectively. Discrimination limits were set at 212 for the green channel and 190 for the blue channel, achieving sensitivities of 100% and 90.9%, and specificities of 62.5% and 60%, respectively, for identifying bilirubin levels above 205 μmol/L. Another study by Kawano et al.[27] proposed a non-invasive method using RGB color and threshold values to isolate facial features for jaundice detection. They measured skin hue and color dispersion, computing eigenvalues of the variance–covariance matrix to identify jaundice. Despite the straightforward approach, the method faced challenges in accurately extracting skin areas, particularly around the eyes and eyebrows, and was limited to a sample size of 64 infants. Juliastuti et al.[28] developed a method for assessing neonatal jaundice through skin color image analysis. They preprocessed images using median filtering and color correction with a reference card, followed by segmentation via K-Means clustering. The analysis was conducted in RGB, HSV, and YCbCr color spaces, generating histograms for statistical parameter extraction. Their regression model demonstrated a correlation with bilirubin levels, achieving a multiple-R of 0.95 during testing, and correctly identified risk zones in 84% of cases. Hashim et al.[29] investigated neonatal jaundice diagnosis through a MATLAB-based graphical user interface. They analyzed images using various color models, including RGB, HSV, and YCbCr, and reported promising results. However, the study's limitations included a small sample size and lack of consideration for different lighting conditions. Ali et al.[30] created an algorithm for diagnosing hereditary jaundice conditions. It used violet transform for image analysis and histogram calculations, employing fuzzy modeling for classification. This method achieved 95–100% accuracy in identifying conditions like Dubin-Johnson and Gilbert syndromes. Mansor et al.[31] proposed a non-invasive method for neonatal jaundice detection using a database of 120 infant images with diverse ethnicities and illumination conditions. They utilized the YCbCr and HSV color spaces for skin detection, discarding luminance to reduce false detections. Statistical texture features such as mean, standard deviation, skewness, kurtosis, energy, and entropy were extracted from the images. Using a k-NN classifier, they achieved an overall classification accuracy of over 90%, with kurtosis providing the highest accuracy of 96%.

Studies have utilized images of different body parts, including the face[31], forehead[15,26,32], sternum[15,23,32,33], abdomen[25,32]. Some research focused on the sclera of the eye for jaundice diagnosis[23,24]. Feature extraction methods varied, including mean, standard deviation, skewness, kurtosis, energy, and entropy[34,35], as well as color spaces like YCbCr and LAB[15,23], RGB[26,35], and HSV[32]. Machine learning algorithms used for bilirubin estimation included k-nearest neighbor (k-NN)[32], support vector regression (SVR), least angle regression (LARS), LARS-Lasso, elastic network, and random forest (RF)[32].

Deep learning has emerged as a groundbreaking approach in medical diagnostics, offering unparalleled capabilities in addressing limitations of traditional methods[36]. Its ability to learn intricate, non-linear patterns

from high-dimensional data makes it especially suited for image-based diagnostic tasks. In the context of neonatal jaundice, deep learning provides transformative potential by enabling precise and non-invasive bilirubin level estimation from neonatal images. Unlike conventional methods such as invasive serum bilirubin testing or subjective visual assessments, deep learning models can analyze complex image features with high accuracy, overcoming challenges related to variability in skin tone, lighting conditions, and image quality[37]. This integration of advanced image processing with deep learning offers a scalable and reliable solution for early detection and management of neonatal jaundice, aiming to improve clinical outcomes while reducing dependency on invasive procedures, particularly in resource-limited settings.

Although neonatal jaundice remains a widespread concern, with numerous studies focused on its diagnosis and management, existing approaches often rely on conventional methods such as invasive blood tests or subjective visual assessments, both of which present limitations in accuracy, accessibility, and patient comfort. While some studies have explored non-invasive techniques using image processing, many are limited by factors such as small sample sizes, reliance on specific ethnic groups, or inadequate handling of variables like skin tone and lighting conditions. Furthermore, fewer studies have explored the potential of deep learning techniques.

This study aims to develop a non-invasive method for estimating and classifying neonatal jaundice using image processing and machine learning. By converting neonatal images into multiple color channels and applying specialized 1D Convolutional Neural Network (1DCNNs) for regression, we aim to predict bilirubin levels accurately. The ultimate goal is to improve early detection and management of neonatal jaundice, reducing reliance on invasive procedures and improving clinical outcomes for newborns. This paper presents an approach for estimating bilirubin levels in neonates using image processing and deep learning techniques. Our main contributions are as follows:

- *Novel 1DCNN framework* We propose a 1DCNN for non-invasive bilirubin level prediction using infant skin images.
- *Color space integration* Combining RGB with HSV enhances feature extraction, improving the model's ability to detect jaundice-related skin color changes.
- *End-to-end learning* The model autonomously learns features from raw data, eliminating the need for manual feature engineering.
- *Non-invasive solution* The approach provides a painless and scalable alternative to traditional bilirubin measurement methods.
- *Clinical applicability* The model's robustness to skin tone and lighting variations makes it suitable for real-world use, especially in resource-limited settings.

The rest of the paper is organized as follows: Section "Materials and methods" details the methodology for converting color space images into time series data and describes the implementation of 1DCNN for bilirubin level regression and also covers the dataset used, including data collection procedures and preprocessing steps. Section "Results" presents experimental results. Section "Discussion" discusses the implications of the findings, comparing the results with existing literature. Finally, Section "Conclusion" summarizes findings and suggests future research directions.

Materials and methods

In this paper, a 1DCNN is employed to predict bilirubin levels for neonatal jaundice detection. Color channels (RGB, HSV, LAB, and YCbCr) are extracted from infant skin images, and each channel is converted into a time series input for the 1DCNN model, which performs regression analysis to predict bilirubin levels. The model's performance is evaluated using RMSE, MAE, and R^2 score. The schematic representation of the proposed framework is presented in Fig. 1.

Dataset

A dataset comprising 2235 images of 745 infants was sourced from the NeoJaundice dataset at Xuzhou Central Hospital, accessible online[38]. The data are publicity available in https://springernature.figshare.com/articles/dataset/NeoJaundice_Neonatal_Jaundice_Evaluation_in_Demographic_Images/22302559?file=39672982. Each image, with an average size of 567×567 pixels, captures infants' body skin regions, specifically the head, face, and chest. Initial binary labels were assigned based on total serum bilirubin readings obtained from the hospital's health information system, using a threshold of 12.9 mg/dL. These labels were subsequently validated by a senior pediatrician with extensive experience. Figure 2 illustrates samples from both low and high bilirubin levels, indicating the distribution across categories. During image acquisition, a standardized color card was employed to ensure color calibration of the captured skin regions. The left image shows a neonate with a bilirubin level of 14.6 mg/dL at the age of 5 days, indicating elevated bilirubin levels that may require medical attention. The right image displays a neonate with a bilirubin level of 3.9 mg/dL, also at the age of 5 days, which is within the normal range. This comparison highlights the variance in skin color corresponding to different bilirubin levels in neonates of the same age[38]. Table 1. provides an overview of the key numerical attributes in the NeoJaundice dataset, which includes 2,235 images of 745 infants.

Preprocessing

In the preprocessing phase, color space adjustment and white balancing are crucial steps, particularly when dealing with neonatal skin images captured using a color calibration card[25]. White balancing is essential to ensure accurate color representation across images. After white balancing, the next step involves segmenting the skin regions within the images. This segmentation process is critical for isolating the areas of interest and enhancing the accuracy of subsequent analyses. The algorithm begins by loading the image and converting it

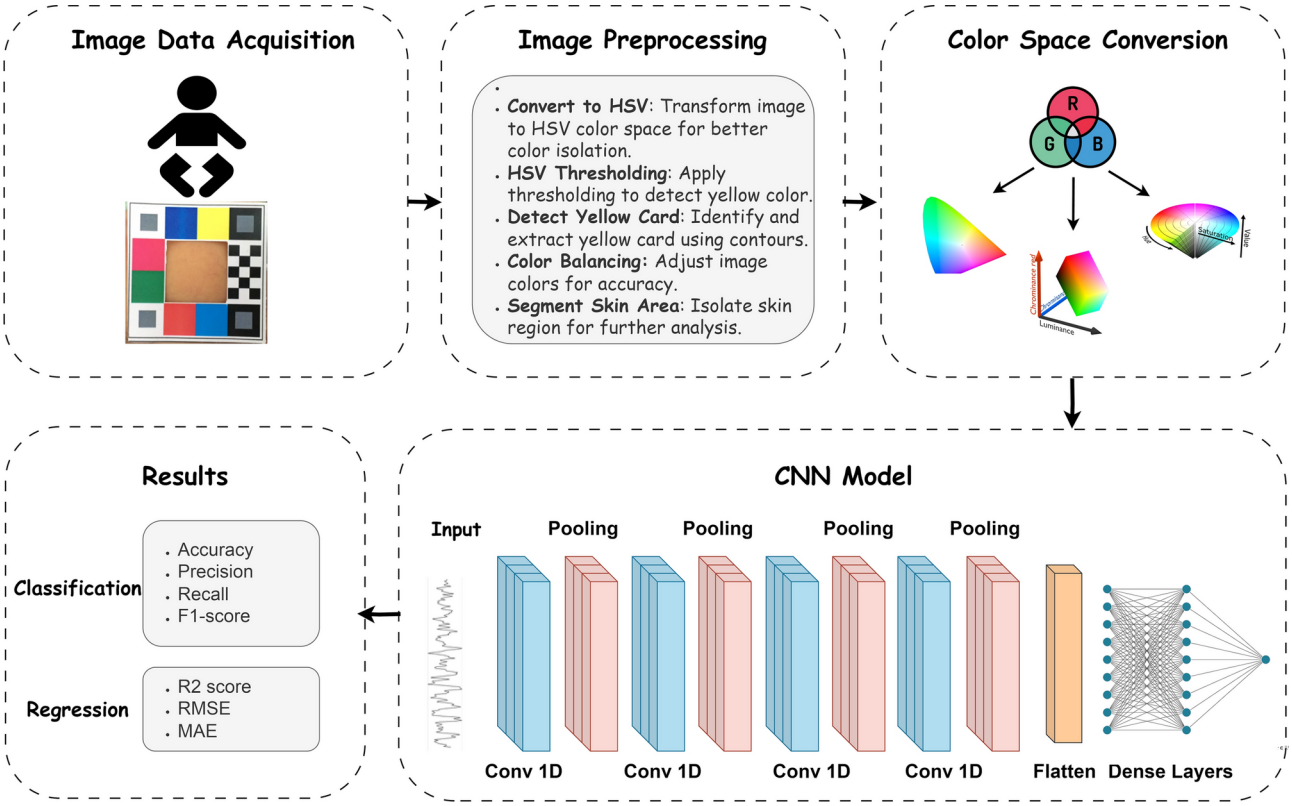


Fig. 1. Diagram outlining the stepwise approach of the proposed framework.

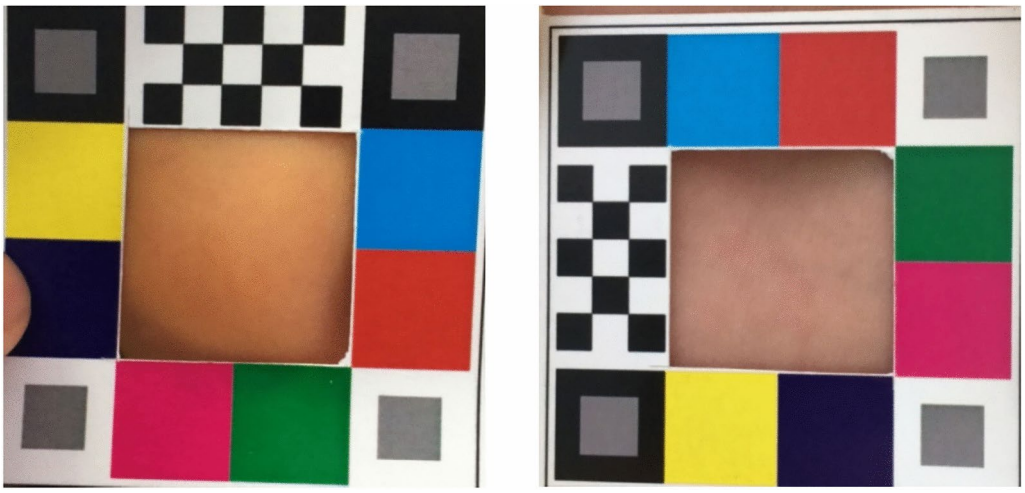


Fig. 2. Left: Neonate, 5 days old, with bilirubin 14.6 mg/dL (abnormal). Right: Neonate, 5 days old, with bilirubin 3.9 mg/dL (normal).

from the BGR to the HSV color space, facilitating easier color thresholding[39]. A yellow color range is defined within the HSV space, allowing for the creation of a mask to isolate yellow regions in the image. This mask is then applied to extract only the yellow areas. Contours of these regions are detected, and the largest contour, assumed to be the yellow reference card, is extracted. The yellow card region is converted to the LAB color space, known for its perceptual uniformity and ability to separate luminance from chromatic components. The average LAB values of the yellow card are computed and used as reference points for adjusting the entire image's color balance. The entire image is converted to the LAB color space, adjusted based on the yellow card's LAB values, and then converted back to the RGB (or BGR) color space. This ensures consistent color correction across the image. For skin segmentation, a fixed coordinate window or Region of Interest (ROI) is defined to manually isolate the skin area[25]. This segmented skin region is displayed to ensure the correct area has been isolated. This

Attribute	Min value	Max value	Mean value
Gestational age	30	42	38.29
Age (day)	0.1	34	6.60
Weight (grams)	1690	5290	3158.78
TSB (mg/dL)	0.0	25.7	11.21
Treatment	0	1	–

Table 1. Summary of numerical attributes in the NeoJaundice dataset, including minimum, maximum, and mean values for gestational age, age, weight, Total Serum Bilirubin (TSB) Levels, and treatment status.

preprocessing algorithm effectively segments the yellow reference card for accurate color balancing and isolates the skin area, enhancing the reliability and accuracy of subsequent analyses.

Convolutional neural network

In this study, we employ a 1DCNN for the estimation of bilirubin levels in neonatal jaundice diagnosis. The 1DCNN architecture is designed to process temporal sequences derived from color channels extracted from infant skin images. This section details the architecture, hyperparameters, and training methodology used for our model.

Model architecture

The 1DCNN architecture consists of several convolutional layers followed by max-pooling layers to extract and downsample features from the input temporal sequences[40]. Subsequently, dense layers are employed to learn higher-level representations and make predictions regarding bilirubin levels. The final layer outputs a single value representing the estimated bilirubin concentration.

- *Convolutional layers* These layers use convolution operations to apply filters across the temporal sequence of color channel data, extracting spatial hierarchies of features relevant to jaundice diagnosis.
- *Max-pooling layers* Following each convolutional layer, max-pooling layers downsample the feature maps to reduce dimensionality while retaining the most salient information.
- *Dense layers* Fully connected dense layers process the flattened output from the convolutional and pooling layers, enabling the model to learn complex patterns and relationships in the data.

In this research, we designed and implemented a 1DCNN for predicting bilirubin levels. The model's architecture is thoughtfully structured to balance complexity and performance, leveraging deep learning capabilities to achieve accurate predictions. The first convolutional layer consists of 32 filters, each with a kernel size of 3, and employs 'same' padding to ensure that the output maintains the same length as the input. The ReLU activation function introduces non-linearity, allowing the model to capture complex patterns in the data. Following this, a MaxPooling1D layer with a pool size of 2 and 'same' padding reduces the dimensionality of the feature map by half, thereby decreasing computational complexity while retaining important features. The second convolutional layer increases the number of filters to 64, maintaining the kernel size at 3 and using 'same' padding. Again, the ReLU activation function is used. This is followed by another MaxPooling1D layer with similar parameters, further reducing the dimensionality. The third convolutional layer further increases the filter count to 128 while keeping the kernel size and padding consistent. The inclusion of another ReLU activation function continues to enable the model to learn non-linear relationships. A MaxPooling1D layer is added to halve the feature map size once more. The fourth convolutional layer, like the third, employs 128 filters with a kernel size of 3 and 'same' padding, along with the ReLU activation function. The subsequent MaxPooling1D layer performs the final down-sampling, reducing the dimensionality of the feature maps. After the convolutional and pooling layers, a Flatten layer is introduced to convert the 1D feature maps into a single 1D vector, making it suitable for input to the fully connected layers. The first fully connected layer contains 256 neurons and uses the ReLU activation function to allow the network to learn intricate representations. To prevent overfitting, a Dropout layer with a rate of 0.3 is applied, randomly setting a fraction of the input units to 0 during training. The second fully connected layer has 128 neurons, again with ReLU activation, to further refine the learned representations. A Dropout layer with a higher rate of 0.5 is applied, providing stronger regularization to mitigate overfitting risks. The final output layer consists of a single neuron with a linear activation function, making it suitable for the regression task of predicting bilirubin levels. This linear activation ensures that the output is a continuous value, aligning with the nature of bilirubin level measurements. Table 2 presents the configuration of each layer in the 1DCNN model.

Hyperparameters and training

Hyperparameters such as filter size, number of filters, dropout rates, and learning rate are optimized through experimentation and validation set performance[41]. The learning rate is initially set and decayed over epochs to facilitate convergence. The number of epochs for training is determined through early stopping criteria to prevent overfitting. The model is trained using Adam optimizer, which adapts the learning rate during training to accelerate convergence and improve performance. Mean Squared Error (MSE) is utilized as the loss function to measure the discrepancy between predicted and actual bilirubin levels. Learning rate scheduling is employed to reduce the learning rate dynamically based on the model's performance on the validation set.

Layer (type)	Output shape	Activation	Param #
Conv1d_1	(None, 3, 32)	ReLU	1572896
Max Pooling1d_1	(None, 2, 32)	–	0
Conv1d_2	(None, 2, 64)	ReLU	6208
Max Pooling1d_2	(None, 1, 64)	–	0
Conv1d_3	(None, 1, 128)	ReLU	24704
Max Pooling1d_3	(None, 1, 128)	–	0
Conv1d_4	(None, 1, 128)	ReLU	49280
Max Pooling1d_4	(None, 1, 128)	–	0
Flatten	(None, 128)	–	0
Dense_1	(None, 256)	ReLU	33024
Dropout_1	(None, 256)	–	0
Dense_2	(None, 128)	ReLU	32896
Dropout_2	(None, 128)	–	0
Dense_3 (output)	(None, 1)	Linear	129

Table 2. Architecture of the 1DCNN model.

Class	Total images	Training set	Validation set	Test set
0–5 mg/dL	345	241	35	69
5–10 mg/dL	519	363	52	104
Above 10 mg/dL	1371	960	137	274
Total	2235	1564	224	447

Table 3. Number of images across dataset splits.

Parameter	Value
Convolutional layers	4
Filter per layer	[32,64,128]
Filter sizer	3
Max-pooling layers	4
Pool size	2
Dense layers	2
Neuron of dense layers	[256, 128]
Dropout rate	0.3, 0.5
Learning rate	Decayed from 0.01
Loss function	Mean Squared Error (MSE)
Optimizer	Adam
Batch size	32
Epochs	150 (early stopping)
Learning rate decay	ReduceLROnPlateau

Table 4. Hyperparameters and training details for the 1DCNN model.

The dataset is partitioned using a hold-out method into training (70%), validation (10%), and test (20%) sets[42]. To ensure robustness of results, the model is trained ten times independently, and the average performance metrics are reported[43]. Each model training involves iterating through a set number of epochs, with early stopping based on validation loss improvement. The Table 3 provides a breakdown of the number of samples for each class in the dataset, including the total number of images and their distribution across the training, validation, and test sets.

The Table 4 summarizes the key hyperparameters and training details used in our 1DCNN model for bilirubin level estimation. This setup ensures the 1DCNN model effectively learns from the temporal color channel sequences extracted from neonatal skin images, enabling accurate prediction of bilirubin levels crucial for diagnosing neonatal jaundice. The models were trained on the premium version of Google Colab, using the Nvidia A100 GPU, which has 40 GB of GPU RAM, and a system with 84 GB of RAM. The deep learning models were implemented using the open-source Keras library, a high-level framework built on TensorFlow.

The integration process of features extracted from two distinct color spaces, RGB and HSV, into a single CNN model is illustrated in the following steps, as shown in Fig. 3:

Input images The original neonatal images are first converted into two different color spaces: RGB and HSV.

Feature extraction Distinct 1DCNN models are employed to extract features from each color space individually, generating two separate sets of feature maps.

Feature concatenation The feature maps from both color spaces are concatenated to form a unified feature vector. This integration leverages the complementary information from both color spaces to improve the model's prediction accuracy.

Integrated CNN model The concatenated feature vector is then fed into a final fully connected model designed for the prediction and classification of neonatal jaundice.

Evaluation metrics

Regression evaluation metrics

- *Mean absolute error (MAE)* Measures the average magnitude of errors in a set of predictions, without considering their direction.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{y}_i\right|$$

- *Root mean square error (RSME)* Provides a measure of the average magnitude of error with the same units as the target variable.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2}$$

- *Coefficient of determination (R-squared)* Indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \overline{y}_i\right)^2}$$

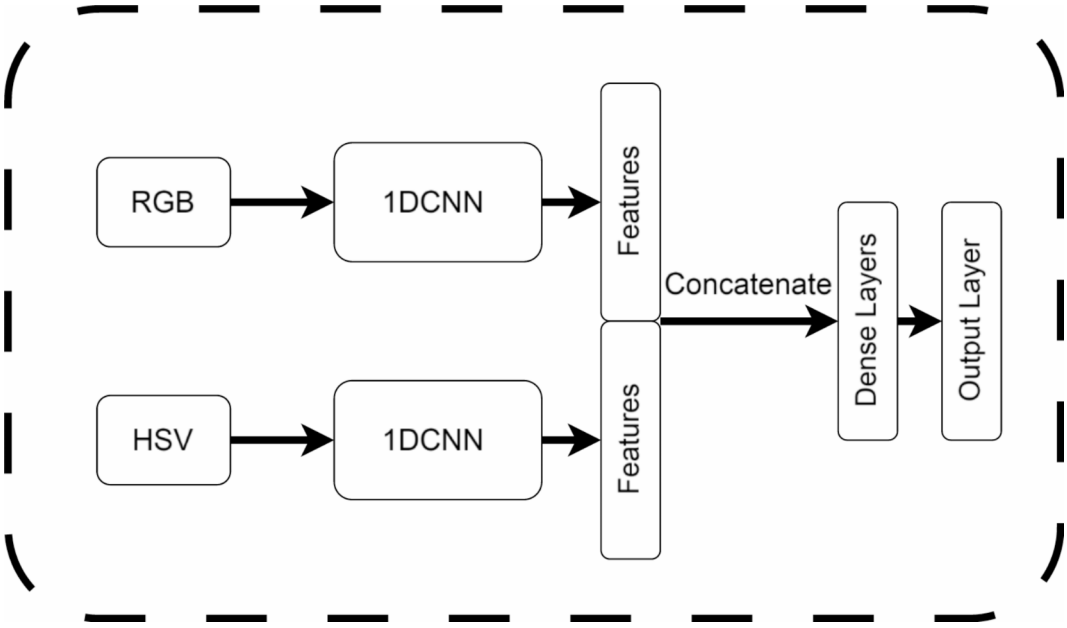


Fig. 3. How to integrate the obtained features with the CNN model. For example, two color spaces RGB and HSV are given.

Classification evaluation metrics

- *Accuracy* Measures the proportion of correct predictions among the total number of predictions made.

$$ACC = \frac{TP + TN}{(TP + TN + FP + FN)}$$

- *Precision* Measures the proportion of true positive predictions (correctly predicted positive instances) among the total predicted positive instances.

$$PRE = \frac{TP}{TP + FP}$$

- *Recall* Measures the proportion of true positive predictions among the actual positive instances.

$$REC = \frac{TP}{TP + FN}$$

- *F1-score* Harmonic mean of precision and recall, providing a single metric that balances both.

$$F - score = 2 \times \frac{PRE \times REC}{PRE + REC}$$

Results

The difference in the values of each color channel in RGB, HSV, LAB, YCbCr spaces for a 4-day-old baby with a normal bilirubin level of 7 and a 3-day-old jaundiced baby with a bilirubin level of 17.1 is shown in Fig. 3. For better display, the first 500 pixels of the image are used to plot the values in Fig. 4. As it is known, the color blue (B) in the RGB color space has changed more between the two babies than red and green. The blue color for a baby who had jaundice has a smaller amount compared to a baby who did not have jaundice. Saturated color (S) in the HSV color space, color (B) in the LAB color space, and color (Cr) in the YCbCr color space had higher values for the baby with jaundice than the baby whose bilirubin was normal. Color (Cb) in the YCbCr color space has a lower average value in nineteen with jaundice.

The performance of the 1DCNN model across different color spaces was evaluated using regression metrics. Table 5 provides a detailed comparison of these metrics across various color spaces and their combinations. The RGB color space achieved a respectable performance with an RMSE of 1.20 and an R^2 score of 0.88, indicating that it can reasonably predict bilirubin levels. The HSV color space showed a slightly better RMSE of 1.17 but had a lower MAE of 1.06 and an R^2 score of 0.86. This suggests that while it might be slightly more precise in some predictions, it is less consistent overall. The LAB color space performed the worst among the individual color spaces, with the highest RMSE of 2.31 and MAE of 1.97, reflecting its lower accuracy and reliability in predicting bilirubin levels. The YCbCr color space showed moderate performance with an RMSE of 1.39 and an R^2 score of 0.87, which is slightly lower than RGB.

When analyzing the combined color spaces, the combination of RGB and HSV color spaces yielded the best performance with the lowest RMSE of 1.13 and the highest R^2 score of 0.91. This combination appears to capture complementary features that enhance prediction accuracy. The combination of RGB and LAB did not perform as well as RGB and HSV, with an RMSE of 1.35 and an R^2 score of 0.82, indicating that adding LAB to RGB did not significantly improve the model's performance. The combination of RGB and YCbCr demonstrated improved performance over RGB alone, with an RMSE of 1.21 and an R^2 score of 0.85, suggesting some beneficial interaction between the features of these two color spaces. The combination of HSV and LAB resulted in an RMSE of 1.58 and an R^2 score of 0.86. Although the R^2 score value is similar to that of individual HSV, the RMSE indicates less precision in this combined approach. The HSV and YCbCr combination had an RMSE of 1.56 and an R^2 score of 0.82, similar to the performance of HSV and LAB, showing that YCbCr did not add significant value to HSV. The combination of LAB and YCbCr performed the worst among the combined color spaces, with an RMSE of 1.78 and an R^2 score of 0.78.

The analysis indicates that combining certain color spaces can improve the model's performance in predicting bilirubin levels. Specifically, the combination of RGB and HSV color spaces provides the best results, suggesting that these two color spaces capture the most relevant features for bilirubin level prediction. The results also highlight the limitations of the LAB color space, both individually and in combination, as it consistently showed lower performance metrics. This suggests that LAB may not effectively represent the key features needed for accurate estimation of bilirubin levels in neonatal images.

To evaluate the performance of our proposed 1DCNN model for bilirubin estimation, we utilized RMSE, MAE, and R^2 as key metrics. RMSE provides a measure of the model's overall prediction error, emphasizing larger errors more heavily, which is critical in medical applications where extreme misestimations can have severe consequences. MAE, on the other hand, measures the absolute error without squaring, offering a more interpretable estimate of average deviation. R^2 quantifies how well the model explains the variance in bilirubin levels, with higher values indicating stronger predictive capabilities. The results indicate that the RGB + HSV model achieved the lowest RMSE (1.13) and MAE (0.89), while also obtaining the highest R^2 score (0.91), demonstrating that the integration of complementary color spaces enhances both precision and generalizability.

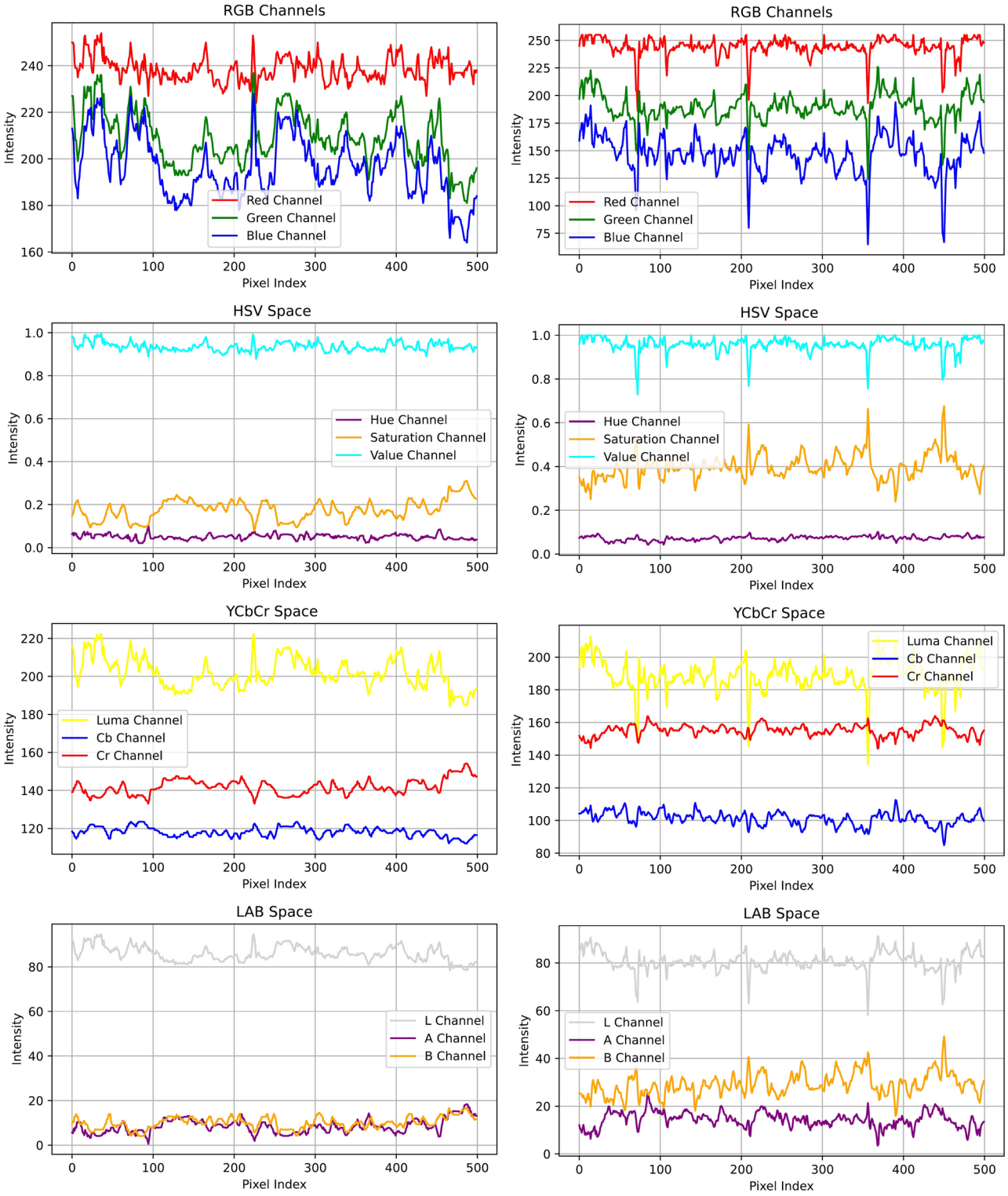


Fig. 4. The values of each pixel in 4 color spaces show the images on the left of a 4-day-old normal baby with a bilirubin of 7 and on the right of a 3-day-old hyperbilirubin baby with a TSB of 17.1.

Based on the promising results obtained from the regression analysis, where the combination of RGB and HSV color spaces yielded the best performance for predicting bilirubin levels, we extended this approach to a classification problem. The aim is to classify neonatal jaundice into three categories based on bilirubin levels: 0–5 mg/dL, 5–10 mg/dL, and greater than 10 mg/dL. We used the same 1DCNN model and combined the RGB and HSV color space features. The model achieved an accuracy of 96.87%, precision of 94.75%, recall of 92.69%, and an F1 score of 93.72%.

Color spaces	RMSE	MAE	R2-Score	Rank
RGB	1.20	0.92	0.88	2
HSV	1.17	1.06	0.86	4
LAB	2.31	1.97	0.84	6
YCbCr	1.39	1.11	0.87	3
RGB + HSV	1.13	0.89	0.91	1
RGB + LAB	1.35	1.08	0.82	7
RGB + YCbCr	1.21	1.14	0.85	5
HSV + LAB	1.58	1.32	0.86	4
HSV + YCbCr	1.56	1.39	0.82	7
LAB + YCbCr	1.78	1.45	0.78	8

Table 5. Performance evaluation of the model across various color space combinations, including RMSE, MAE, R^2 Score, and ranking.

Color spaces	Accuracy	Precision	Recall	F1 score
RGB	89.33	91.62	86.22	87.79
HSV	91.44	92.99	90.24	90.43
LAB	89.21	93.62	94.44	87.73
YCbCr	89.61	89.61	91.29	90.19
RGB + HSV	96.87	94.75	92.69	93.72
RGB + LAB	92.92	87.18	89.61	90.69
RGB + YCbCr	91.84	89.07	87.48	88.17
HSV + LAB	86.41	88.19	90.81	87.55
HSV + YCbCr	87.16	91.26	90.57	88.52
LAB + YCbCr	89.36	90.12	88.26	86.18

Table 6. Performance evaluation of the model across various color space combinations, including accuracy, precision, recall, and F1 score.

The results of the classification task for different color spaces and their combinations are presented in Table 6. The metrics reported are based on the weighted average for the three classes. Among individual color spaces, HSV achieved the highest accuracy (91.44%), precision (92.99%), recall (90.24%), and F1 score (90.43%), indicating its effectiveness in capturing relevant features for classification. RGB and LAB also performed well, with accuracies of 89.33% and 89.21%, respectively. Notably, LAB achieved the highest recall (94.44%) among all individual color spaces, suggesting its capability to detect positive cases effectively.

When combining color spaces, RGB + HSV outperformed all others, achieving the highest accuracy of 96.87%, precision of 94.75%, recall of 92.69%, and F1 score of 93.72%, reflecting a strong synergy between these two color spaces. Other combinations, such as RGB + LAB and RGB + YCbCr, achieved accuracies of 92.92% and 91.84%, respectively, with respectable precision and recall values. On the other hand, combinations like HSV + LAB and HSV + YCbCr yielded slightly lower accuracies of 86.41% and 87.16%, respectively.

The accuracy of 96.87% underscores the model's capability to correctly classify the vast majority of samples, affirming the effectiveness of the training process and the underlying architecture. Precision, at 94.75%, signifies that the model has a low rate of false positives, meaning that it is highly accurate when it predicts a positive outcome. Recall, recorded at 92.69%, demonstrates the model's efficacy in identifying true positives from the dataset. Figure 5 illustrates the confusion matrix corresponding to the best classification results achieved by our model. The matrix categorizes the predictions across three blood bilirubin levels: 0–5, 5–10, and 10 > .

To ensure the robustness and consistency of our model, we employed a tenfold cross-validation strategy during testing. This technique is highly regarded in deep learning research as it allows for comprehensive assessment of the model's performance across diverse data partitions, minimizing potential biases that may arise from data splitting methods. The outcomes of these tenfold cross-validation experiments are presented in Fig. 6.

Discussion

The results presented in Table 7 underscore the significant advancements achieved in this study compared to prior works in the domain of neonatal bilirubin estimation. Previous studies have predominantly employed statistical feature extraction combined with machine learning algorithms or regression models, with varying degrees of success. To further substantiate these advancements, a comparative analysis of evaluation metrics provides deeper insight into the effectiveness of different color space combinations. While RMSE and MAE both quantify model errors, RMSE disproportionately penalizes larger deviations, making it particularly useful in ensuring that the model does not produce extreme mispredictions. The RGB + HSV model's lowest RMSE (1.13) and MAE (0.89) confirm its stability and accuracy in bilirubin estimation. Additionally, the R^2 value of 0.91 indicates that

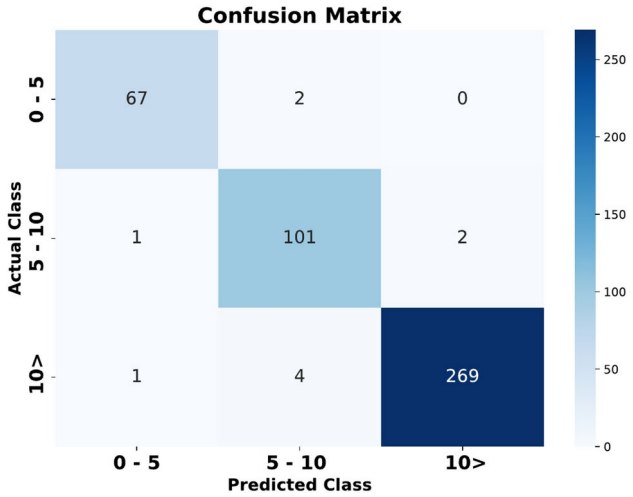


Fig. 5. Confusion matrix of best classification result across three classed.

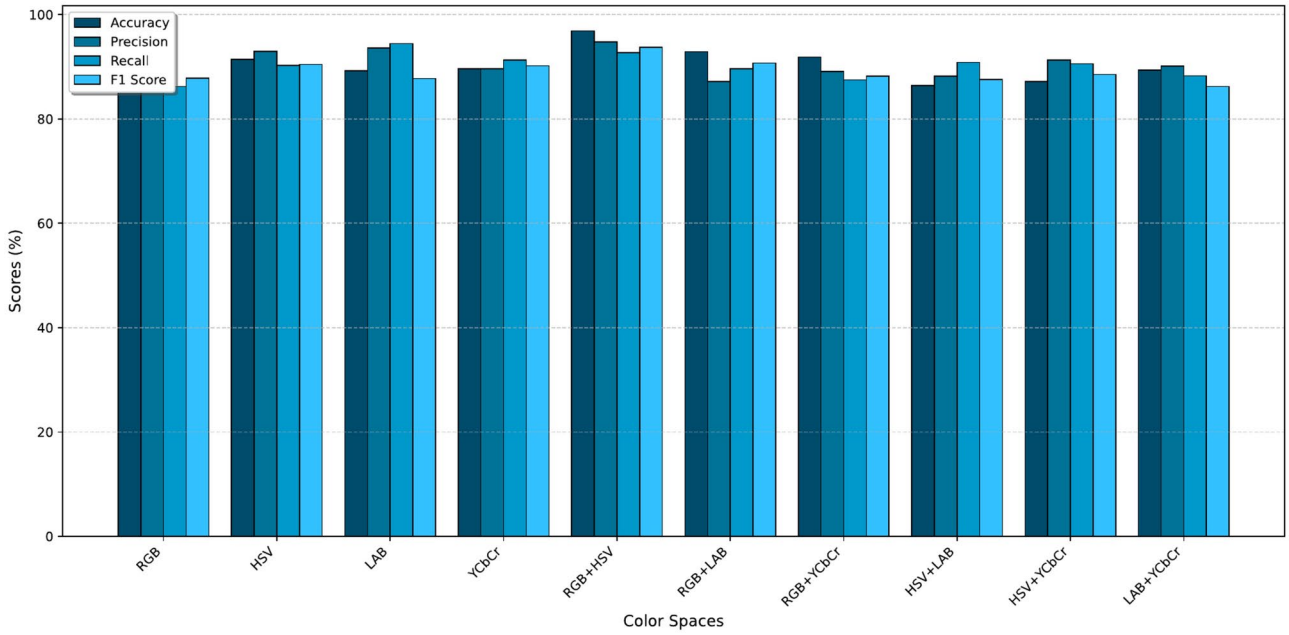


Fig. 6. Comparison of performance metrics (Accuracy, Precision, Recall, and F1 Score) for bilirubin level classification across different color spaces.

the model effectively captures the variance in bilirubin levels, demonstrating its robustness. Notably, while some models achieved relatively low RMSE values, their lower R^2 scores suggest weaker generalization to unseen data. This highlights the importance of considering all three metrics together to ensure both precision and reliability in medical AI applications. Below, we provide a comparative analysis of these methodologies to underscore the advantages of our approach.

A number of studies, including those by Mansor et al.[31] and Laddi et al.[44], utilized statistical features in conjunction with classifiers such as KNN and ANFIS, reporting accuracies between 90 and 96%. While these methods are computationally efficient, they inherently depend on the quality and relevance of manually extracted features. Such dependency limits their ability to generalize across larger and more diverse datasets. In contrast, the proposed 1DCNN in this study bypasses the need for manual feature engineering, autonomously extracting patterns from raw input data. This capability is reflected in the superior accuracy of 96.87%, a notable improvement over traditional methods.

Regression techniques, as employed by De Greef et al.[15] (ensemble of KNNR, SVR, and others) and Leung et al.[45] (linear regression), achieved correlation coefficients of 0.85 and 0.75, respectively. While regression-based models are effective for continuous variable prediction, their reliance on linear or predefined relationships constrains their predictive power, particularly for datasets with nonlinear complexities. The proposed 1DCNN,

References	Data	Method	Results
Mansor et al.[31]	120 images	Statistical features, KNN classifier	Accuracy = 90–96%
Laddi et al.[44]	420 images	Statistical features, PCA, ANFIS	Accuracy = 90%
De Greef et al.[15]	100 images	Statistical features, regression algorithm: ensemble of KNNR, SVR, LAR, ENR, RFR	Correlation = 0.85
Leung et al.[45]	110 images	Statistical features, linear regression	Correlation = 0.75
Aydin et al.[25]	80 images	KNN, SVM	Accuracy = 75–85%
Outlaw et al.[34]	86 images	Statistical features, Regression	Correlation = 0.71
Swarna et al.[32]	35 images	Statistical features, ensemble learning regression	Correlation = 0.55–0.6
Chakraborty et al.[46]	–	Convolutional neural network	Accuracy = 84%
Althnian et al.[47]	100 images	VGG16, SVM, MLP, DT, RF	Accuracy = 86.83%
Yuan et al.[48]	3465 images	Statistical features, DT, KNN, SVM, RF and LightGBM	Accuracy = 84.80%
This study	2235 images	1DCNN	Accuracy = 96.87%
			RMSE = 1.13
			R^2 Score = 0.91

Table 7. Comparison of methods and results for bilirubin level estimation across previous studies and the proposed approach.

with its inherent ability to model nonlinearity, achieves an R^2 score of 0.91, demonstrating superior explanatory power. Additionally, the model's low RMSE of 1.13 further confirms its precision, outperforming regression approaches used in previous studies.

Several prior works, such as those by Aydin et al.[25] and Swarna et al.[32], relied on datasets comprising fewer than 100 images, leading to reduced generalizability and lower predictive performance (accuracies ranging from 75 to 85%, and correlations as low as 0.55–0.6). Small dataset sizes inherently limit the model's ability to capture inter-subject variability and complex patterns in bilirubin levels. This study, by leveraging a considerably larger dataset of 2235 images, addresses this limitation and demonstrates the robustness of the proposed method in handling diverse data distributions.

Recent advances in deep learning and ensemble techniques have shown promise for bilirubin level estimation. Chakraborty et al.[46] utilized CNNs, achieving an accuracy of 84%, while Althnian et al.[47] applied VGG16 and machine learning classifiers such as SVM and RF, reporting an accuracy of 86.83%. While these studies highlight the potential of deep learning, their reliance on relatively small datasets (e.g., 100 images in Althnian et al.[47]) limits their applicability in real-world scenarios. Additionally, ensemble approaches, such as the one employed by Yuan et al.[48] using methods like DT, RF, and LightGBM, achieved an accuracy of 84.80%. Despite their effectiveness, these methods are still dependent on manual feature extraction. The proposed 1DCNN, by learning end-to-end representations from a larger dataset, not only outperforms these methods in terms of accuracy but also offers a scalable and generalizable solution.

Advantages of the proposed approach are as follows:

- *Superior predictive performance* The proposed 1DCNN achieved the highest accuracy of 96.87%, outperforming both traditional machine learning and deep learning methods. Its capacity to autonomously learn relevant features from raw data significantly contributes to this improvement.
- *Enhanced regression metrics* The study reports the lowest RMSE (1.13) and the highest R^2 score (0.91) among all reviewed works, showcasing the model's robustness and precision in predicting bilirubin levels.
- *Overcoming dataset limitations* Unlike prior studies that relied on limited datasets, this study utilized a dataset of 2235 images, enabling the model to capture a wider range of variability in bilirubin levels. This scalability ensures broader applicability in clinical scenarios.
- *Elimination of handcrafted features* While earlier studies relied heavily on manually extracted statistical features, the 1DCNN's ability to automatically learn complex, hierarchical features eliminates this dependency, reducing human intervention and potential biases.
- *Real-World Applicability* The combination of high accuracy, robust regression metrics, and scalability makes the proposed method highly suitable for integration into real-world neonatal healthcare systems, potentially improving early diagnosis and intervention strategies.

This study has several limitations that should be considered. First, the dataset used was relatively small and may not fully capture the diversity of neonatal skin tones and bilirubin levels, limiting the model's generalizability. The controlled environment of the data collection also does not reflect real-world clinical settings where lighting variations and image quality can differ, potentially affecting the model's performance. Future research should focus on expanding the dataset to include a wider range of skin tones, environmental factors, and bilirubin levels, enhancing the model's applicability in diverse clinical scenarios. Investigating advanced architectures, such as 2D or 3D CNNs, transformer models, or attention mechanisms, may lead to more robust feature extraction and better performance. Lastly, deploying the model for real-time analysis in clinical settings, particularly through mobile or wearable devices, could pave the way for widespread, non-invasive neonatal jaundice screening in both hospitals and home environments.

Conclusion

This research demonstrates the effectiveness of utilizing one-dimensional convolutional neural network for the non-invasive prediction and classification of neonatal jaundice. By leveraging the unique features of various color spaces specifically RGB, HSV, LAB, and YCbCr. we have shown that the integration of RGB and HSV color spaces yields the most accurate and reliable predictions. This is evidenced by the combination's achievement of the lowest RMSE of 1.13 and the highest R^2 score of 0.91. These metrics indicate a high degree of accuracy and reliability in estimating bilirubin levels, which is crucial for the early detection and management of neonatal jaundice.

Overall, this study provides a solid foundation for the continued development of non-invasive diagnostic tools for neonatal jaundice, paving the way for improved neonatal care and better health outcomes. The integration of advanced deep learning techniques with innovative feature extraction from various color spaces represents a significant step forward in medical imaging and diagnostics.

Despite these promising results, this study has several limitations that warrant further investigation. One limitation is that the model's performance has been evaluated using only a single publicly available dataset. Another limitation is that the current model relies solely on skin color analysis without considering other potential biomarkers, such as eye sclera discoloration, which could provide additional diagnostic information. Furthermore, our approach is currently limited to a single deep learning architecture 1DCNN.

To address these limitations and build upon our findings, several future directions can be pursued. Expanding the dataset to include a broader range of neonatal skin tones, age groups, and clinical conditions would enhance the model's generalizability and applicability across diverse populations. Investigating advanced deep learning architectures, such as 2D CNNs, Vision Transformers (ViTs), and attention-based models, could further improve feature extraction from multi-color space representations, leading to more robust and accurate predictions. Additionally, developing a real-time mobile or cloud-based implementation of the proposed method would facilitate clinical deployment, enabling non-invasive jaundice screening in hospitals and remote settings. Integrating additional physiological parameters, such as gestational age and weight, alongside image-based features, may also improve predictive performance by providing a more comprehensive assessment of neonatal jaundice risk. Finally, conducting prospective clinical validation studies in real-world hospital environments is essential to evaluate the model's robustness under varying lighting conditions and imaging devices, ensuring its effectiveness in practical medical applications.

Data availability

The data are publicity available in https://springernature.figshare.com/articles/dataset/NeoJaundice_Neonatal_Jaundice_Evaluation_in_Demographic_Images/22302559?file=39672982.

Received: 4 October 2024; Accepted: 26 March 2025
Published online: 04 April 2025

References

1. Olusanya, B. O., Kaplan, M. & Hansen, T. W. Neonatal hyperbilirubinaemia: A global perspective. *Lancet Child Adolesc. Health* **2**(8), 610–620 (2018).
2. Maheshwari, V., Díaz-González de Ferris, M. E., Filler, G. & Kotanko, P. Novel extracorporeal treatment for severe neonatal jaundice: A mathematical modelling study of allo-hemodialysis. *Sci. Rep.* **14**(1), 21910 (2024).
3. Rennie, J., Burman-Roy, S. & Murphy, M. S. Neonatal jaundice: Summary of NICE guidance. *Bmj* **340** (2010).
4. Kirk, J. M. Neonatal jaundice: A critical review of the role and practice of bilirubin analysis. *Ann. Clin. Biochem.* **45**(5), 452–462 (2008).
5. van der Geest, B. A. et al. Assessment, management, and incidence of neonatal jaundice in healthy neonates cared for in primary care: a prospective cohort study. *Sci. Rep.* **12**(1), 14385 (2022).
6. Singla, R. & Singh, S. A framework for detection of jaundice in new born babies using homomorphic filtering based image processing. In *2016 International Conference on Inventive Computation Technologies (ICICT)* Vol. 3, 1–5 (IEEE, 2016).
7. Cohen, R. S., Wong, R. J. & Stevenson, D. K. Understanding neonatal jaundice: a perspective on causation. *Pediatr. Neonatol.* **51**(3), 143–148 (2010).
8. Halder, A. et al. A novel whole spectrum-based non-invasive screening device for neonatal hyperbilirubinemia. *IEEE J. Biomed. Health Inform.* **23**(6), 2347–2353 (2019).
9. Ahmadpour-Kacho, M. et al. Cord blood alkaline phosphatase as an indicator of neonatal jaundice. *Iran. J. Pediatrics* **25**(5) (2015).
10. Seidman, D. S. et al. Neonatal hyperbilirubinemia and physical and cognitive performance at 17 years of age. *Pediatrics* **88**(4), 828–833 (1991).
11. Boskabadi, H., Zakerihamidi, M., Moradi, A. & Bakhshaee, M. Risk factors for sensorineural hearing loss in neonatal hyperbilirubinemia. *Iran. J. Otorhinolaryngol.* **30**(99), 195 (2018).
12. Tsao, P.-C. et al. Long-term neurodevelopmental outcomes of significant neonatal jaundice in Taiwan from 2000–2003: A nationwide, population-based cohort study. *Sci. Rep.* **10**(1), 11374 (2020).
13. Porter, M. L. & Dennis, B. L. Hyperbilirubinemia in the term newborn. *Am. Fam. Phys.* **65**(4), 599–607 (2002).
14. Riskin, A., Tamir, A., Kugelman, A., Hemo, M. & Bader, D. Is visual assessment of jaundice reliable as a screening tool to detect significant neonatal hyperbilirubinemia?. *J. Pediatrics* **152**(6), 782–787 (2008).
15. De Greef, L. et al. Bilicam: Using mobile phones to monitor newborn jaundice. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing* 331–342 (2014).
16. Kesarwani, A., Das, S., Dalui, M. & Kisku, D. R. Non-invasive haemoglobin estimation using different colour and texture features of palm. In *International Conference on Computer Vision and Image Processing* 176–189 (Springer, Cham, 2022).
17. Sanpavat, S. & Nuchprayoon, I. Transcutaneous bilirubin in the pre-term infants. *J.-Med. Assoc. Thailand* **90**(9), 1803 (2007).
18. Rahmawati, D., Sampurna, M. T. A., Etika, R., Utomo, M. T. & Bos, A. F. Transcutaneous bilirubin level to predict hyperbilirubinemia in preterm neonates. *F1000Research* **9** (2020).
19. Boo, N. Y. & Ishak, S. Prediction of severe hyperbilirubinaemia using the Bilicheck transcutaneous bilirubinometer. *J. Paediatr. Child Health* **43**(4), 297–302 (2007).
20. Engle, W. D., Jackson, G. L., Stehel, E. K., Sendelbach, D. M. & Manning, M. D. Evaluation of a transcutaneous jaundice meter following hospital discharge in term and near-term neonates. *J. Perinatol.* **25**(7), 486–490 (2005).

21. Dam-Vervloet, A. J. et al. Skin color influences transcutaneous bilirubin measurements: A systematic in vitro evaluation. *Pediatric Res.* 1–5 (2024).
22. Taylor, J. A. et al. Discrepancies between transcutaneous and serum bilirubin measurements. *Pediatrics* **135**(2), 224–231 (2015).
23. Taylor, J. A. et al. Use of a smartphone app to assess neonatal jaundice. *Pediatrics* **140**(3) (2017).
24. Abdulrazzak, A. Y., Mohammed, S. L., Al-Naji, A. & Chahl, J. Neonatal jaundice detection using machine-learning algorithms: A comparative study. In *AIP Conference Proceedings* Vol. 3232, no. 1 (AIP Publishing, 2024).
25. Aydın, M., Hardalaç, F., Ural, B. & Karap, S. Neonatal jaundice detection system. *J. Med. Syst.* **40**, 1–11 (2016).
26. Munkholm, S. B., Krøgholt, T., Ebbesen, F., Szecsi, P. B. & Kristensen, S. R. The smartphone camera as a potential method for transcutaneous bilirubin measurement. *PLoS ONE* **13**(6), e0197938 (2018).
27. Kawano, S., Zin, T. T. & Kodama, Y. A study on non-contact and non-invasive neonatal jaundice detection and bilirubin value prediction. In *2018 IEEE 7th Global Conference on Consumer Electronics (GCCE)* 401–402 (IEEE, 2018).
28. Juliastuti, E., Nadhira, V., Satwika, Y. W., Aziz, N. A. & Zahra, N. Risk zone estimation of newborn jaundice based on skin color image analysis. In *2019 6th International Conference on Instrumentation, Control, and Automation (ICA)* 176–181 (IEEE, 2019).
29. Hashim, W., Al-Naji, A., Al-Rayahi, I. A. & Oudah, M. Computer vision for jaundice detection in neonates using graphic user interface. In *IOP Conference Series: Materials Science and Engineering* vol. 1105, no. 1, 012076 (IOP Publishing, 2021).
30. Ali, S. K., Zou, B. & Ali, A. An algorithm for diagnosis of the three kinds of constitutional jaundice. *Int. Arab J. Inf. Technol.* **7**(4), 441–448 (2010).
31. Mansor, M., Hariharan, M., Basah, S. N. & Yaacob, S. New newborn jaundice monitoring scheme based on combination of pre-processing and color detection method. *Neurocomputing* **120**, 258–261 (2013).
32. Swarna, S., Pasupathy, S., Chinnasami, B., Manasa, D. & Ramraj, B. The smart phone study: assessing the reliability and accuracy of neonatal jaundice measurement using smart phone application. *Int. J. Contemp. Pediatr,* **5**(285) (2018).
33. Saini, N. & Kumar, A. Comparison of non-invasive Bilirubin detection techniques for Jaundice prediction. *Int. J. Innov. Res. Electr. Electron. Instrum. Control Eng.* **4**(5), 151–153 (2016).
34. Outlaw, F., Meek, J., MacDonald, L. W. & Leung, T. S. Screening for neonatal jaundice with a smartphone. In *Proceedings of the 2017 International Conference on Digital Health* 241–242 (2017).
35. Castro-Ramos, J. et al. Detecting jaundice by using digital image processing. In *Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXI*, vol. 8949 290–296 (SPIE, 2014).
36. Lanjewar, M. G., Parab, J. S., Shaikh, A. Y. & Sequeira, M. CNN with machine learning approaches using ExtraTreesClassifier and MRMR feature selection techniques to detect liver diseases on cloud. *Clust. Comput.* **26**(6), 3657–3672 (2023).
37. Lanjewar, M. G., Panchbhai, K. G. & Patle, L. B. Fusion of transfer learning models with LSTM for detection of breast cancer using ultrasound images. *Comput. Biol. Med.* **169**, 107914 (2024).
38. Wang, D. et al. A real-world dataset and benchmark for foundation model adaptation in medical image classification. *Sci. Data* **10**(1), 574 (2023).
39. Chernov, V., Alander, J. & Bochko, V. Integer-based accurate conversion between RGB and HSV color spaces. *Comput. Electr. Eng.* **46**, 328–337 (2015).
40. Kiranyaz, S. et al. 1D convolutional neural networks and applications: A survey. *Mech. Syst. Signal Process.* **151**, 107398 (2021).
41. Kilichev, D. & Kim, W. Hyperparameter optimization for 1D-CNN-based network intrusion detection using GA and PSO. *Mathematics* **11**(17), 3724 (2023).
42. Shen, S.C.-Y. & Buehler, M. J. Nature-inspired architected materials using unsupervised deep learning. *Commun. Eng.* **1**(1), 37 (2022).
43. Rainio, O., Teuho, J. & Klén, R. Evaluation metrics and statistical tests for machine learning. *Sci. Rep.* **14**(1), 6086 (2024).
44. Laddi, A., Kumar, S., Sharma, S. & Kumar, A. Non-invasive jaundice detection using machine vision. *IETE J. Res.* **59**(5), 591–596 (2013).
45. Leung, T. S. et al. Screening neonatal jaundice based on the sclera color of the eye using digital photography. *Biomed. Opt. Express* **6**(11), 4529–4538 (2015).
46. Chakraborty, A., Goud, S., Shetty, V. & Bhattacharyya, B. Neonatal jaundice detection system using CNN algorithm and image processing. *Int. J. Electr. Eng. Technol.* **11**(3) (2020).
47. Althnian, A., Almanea, N. & Aloboud, N. Neonatal jaundice diagnosis using a smartphone camera based on eye, skin, and fused features with transfer learning. *Sensors* **21**(21), 7038 (2021).
48. Yuan, Y. et al. Application of machine learning algorithms for accurate determination of bilirubin level on in vitro engineered tissue phantom images. *Sci. Rep.* **14**(1), 5952 (2024).

Author contributions
F.M.: writing and editing manuscript, review the manuscript, software analysis, visualization results.

Funding
No financial benefits have been received.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to F.M.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025